

MechaScreener: Large Language Model-Based Automated Screening for Systematic Reviews and Research

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Abstract

The screening of thousands of articles is the primary bottleneck in the systematic review (SR) process. This paper presents **MechaScreener**, an automated screening tool that utilizes a Large Language Model (LLM) to rank article relevance based on PICOT or general eligibility criteria. Using an LLM, we developed a scoring mechanism (1–5) to triage references. Our evaluation across 15 diverse research libraries demonstrates that MechaScreener achieves a perfect mean recall of 1.00 (100%) while maintaining an overall mean specificity of 0.61. This effectively reduces the manual screening workload by over 60% without missing any relevant studies.

1 Introduction

Systematic reviews (SRs) require a comprehensive search of literature, often resulting in thousands of references that must be manually screened. Automated tools aim to reduce this burden. This study evaluates **MechaScreener**, an automated tool that leverages a Large Language Model (LLM) to evaluate the inclusion probability of references based on specific researcher-defined criteria.

2 Methods

2.1 System Architecture

MechaScreener was developed using a Node.js/TypeScript environment. The system processes references in batches, extracting the title and abstract and pairing them with a user-provided screening criteria string (formatted as PICOT or general inclusion/exclusion rules).

2.2 Prompting and Scoring

The MechaScreener system utilizes a structured prompt consisting of:

1. **System Prompt:** Defined instructions for the LLM to act as a screening assistant.
2. **Reference Data:** Title and abstract of the paper.
3. **Criteria:** The eligibility criteria for the specific review.

The model is instructed to provide a score from 1 (low probability of inclusion) to 5 (high probability of inclusion). We utilized a model with a **temperature** of 0 and **top_k** of 1 to ensure deterministic, reproducible results.

2.3 Experimental Design

The evaluation of MechaScreenener was conducted in two phases:

- **Development Dataset:** Five libraries used to calibrate the prompt and scoring threshold to maximize recall and specificity.
- **Evaluation Dataset:** Ten additional libraries used for independent validation of the tool’s performance across different medical and social science domains (e.g., COVID-19, diabetes, bone health). This includes 5 SRs that were looking for Randomized Controlled Trials (RCTs) and 5 SRs that were looking at non-RCT studies.

Any references missing abstract data were excluded from the evaluation.

2.4 Metrics

The performance of MechaScreenener is evaluated primarily using two metrics: recall and specificity.

Recall (or sensitivity) measures the proportion of actual relevant studies that were correctly identified by the tool. In the context of systematic reviews, maximizing recall is paramount to ensure no critical literature is missed. It is defined as:

$$\text{Recall} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}} \quad (1)$$

Specificity measures the proportion of irrelevant studies that were correctly excluded. A higher specificity indicates a greater reduction in the manual screening workload, as fewer irrelevant articles are passed to the human reviewers. It is defined as:

$$\text{Specificity} = \frac{\text{True Negatives (TN)}}{\text{True Negatives (TN)} + \text{False Positives (FP)}} \quad (2)$$

2.5 Development Results

Five libraries were used for development to maximize recall (number of references correctly included) and specificity (number of references correctly excluded). The developmental results for these libraries are presented below in Table 1. These results are presented in terms of recall and specificity below in Table 2.

Table 1: Results Across 5 Development Libraries

Library Name	No. of Refs	Include Studies Correctly Included	Exclude Studies Correctly Excluded
Antibiotic Prescribing and Tele-health	934	13/13	612/921
Long Covid	1,216	16/16	552/1,200
Natural History Primary Care	5,913	11/11	3,169/5,902
Non Drug Interventions	5,342	35/35	3,972/5,307
Salt Substitution	889	23/23	733/866

2.6 Evaluation Dataset

Ten reference libraries were used for evaluation of performance of Mechascreeener. These ten libraries include a diverse range of topics shown below (5 studies that were RCTs and 5 studies that were non-RCTs).

Table 2: Mean Statistics for MechaScreener Across 5 Development Libraries

Library Name	Recall	Specificity
Antibiotic Prescribing and Telehealth	1.0000	0.6645
Long Covid	1.0000	0.4600
Natural History Primary Care	1.0000	0.5369
Non Drug Interventions	1.0000	0.7484
Salt Substitution	1.0000	0.8464
Mean	1.0000	0.6513

Table 3: Overview of References and Missing Abstracts in Evaluation Libraries

Library Name	Total References	Missing Abstracts
RCT Studies		
Balneotherapy for Chronic Venous	338	11
Calcium Vitamin D for Bones	5,756	255
Methylxanthine for Apnea	2,781	334
Phosphodiesterase 5 Inhibitors	2,672	338
Topical and Oral Steroids for Om	4,292	583
Non-RCT Studies		
Covid Schools	2,162	21
Diabetes Tb	5,177	165
Falls Prevention	18,098	1,095
Fluoride Fluorosis	11,829	2,527
Sanitation Diarrhoea	10,770	297

3 Results

MechaScreener was tested across 10 evaluation libraries containing a total of over 58,000 references. Figure 1 summarizes the performance metrics visually across each library and highlights the overall average for the evaluation set.

Within the evaluation dataset, the system achieved a perfect mean recall score (1.0000), meaning no relevant articles were missed by the tool. The mean specificity was 0.6095, indicating that MechaScreener correctly excluded 61.0% of irrelevant articles, significantly narrowing the pool for human review.

An individual breakdown of each evaluation library is presented in Table 4.

4 Discussion

In the context of SRs, recall is the most critical metric; missing a single relevant study can invalidate a review. Our results show that MechaScreener, when prompted with a structured 1–5 scoring system, can act as an effective "first-pass" filter.

The primary value of MechaScreener lies in its ability to safely remove 60–62% of the noise while guaranteeing that all relevant literature remains in the set for the second stage of screening.

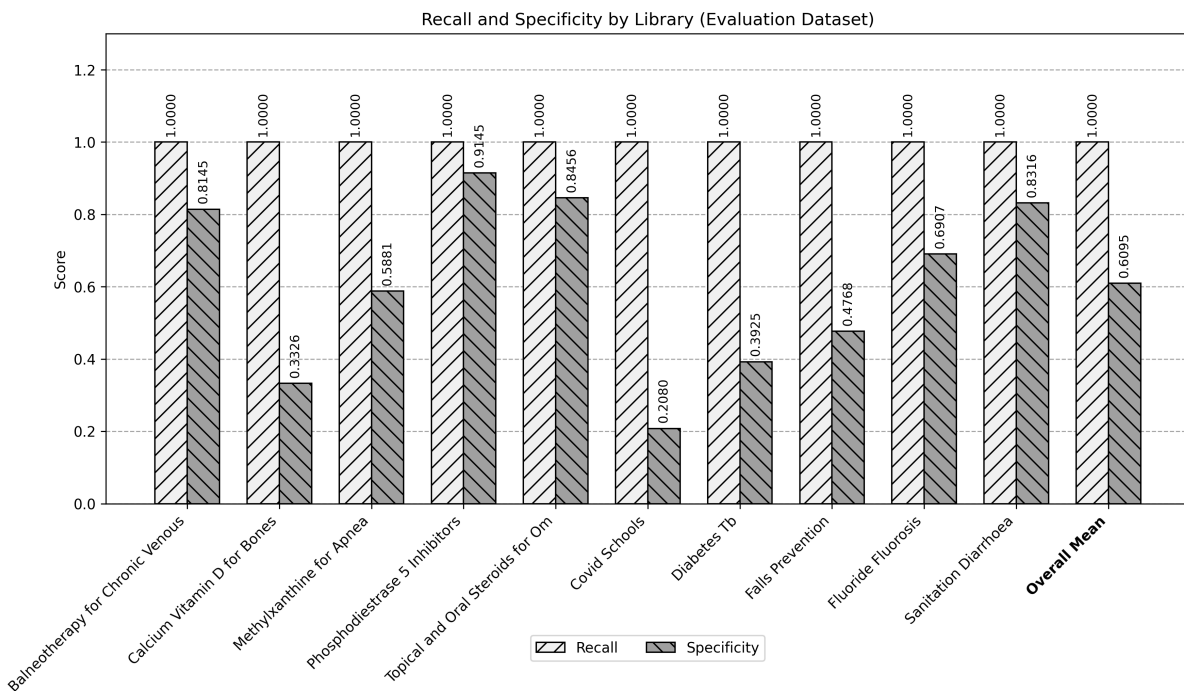


Figure 1: Recall and Specificity across 10 Evaluation Libraries, including the overall Mean.

5 Conclusion

MechaScreener provides a reliable method for screening research articles. By leveraging large-scale LLMs and a probabilistic scoring approach, researchers can use MechaScreener to reduce their screening workload by more than half while maintaining the rigorous standards required for systematic literature synthesis.

Data and Code Availability

The dataset, source code for the analysis, along with the scripts used to generate the figures and tables presented in this paper, are available in the GitHub repository:

<https://github.com/IEBH/mechascreener-paper>

Table 4: Results Across 10 Evaluation Libraries

Library Name	No. of Refs	Include Studies Correctly Included	Exclude Studies Correctly Excluded
Balneotherapy for Chronic Venous	327	9/9	259/318
Calcium Vitamin D for Bones	5,493	6/6	1,825/5,487
Methylxanthine for Apnea	2,446	28/28	1,422/2,418
Phosphodiesterase 5 Inhibitors	2,333	6/6	2,128/2,327
Topical and Oral Steroids for Om	3,706	28/28	3,110/3,678
Covid Schools	2,141	30/30	439/2,111
Diabetes Tb	5,011	48/48	1,948/4,963
Falls Prevention	17,000	9/9	8,102/16,991
Fluoride Fluorosis	9,244	40/40	6,357/9,204
Sanitation Diarrhoea	10,473	31/31	8,684/10,442